

## 2006 Extension 1 Term 4

### QUESTION ONE

### MARKS

- (a) (i) Find  $\int (3x+5)^8 dx$ . 1
- (ii) Use the substitution  $u = 3 - x^2$  to find  $\int \frac{xdx}{\sqrt{3-x^2}}$ . 2
- (b) Express  $\sqrt{12} \sin x + 2\cos x$  in the form of  $A\cos(x + \alpha)$ ; where  $A > 0$  and  $0 \leq \alpha \leq 2\pi$ . 3
- (c) (i) By using one application of Simpson's Rule ( i.e. 3 function values), find an approximation for the area bounded by  $y = \frac{1}{x}$ ,  $1 \leq x \leq 2$ , and the  $x$  axis, leaving your answer as a fraction in simplest terms. 2
- (ii) Using integration, find an expression for the exact area in part (i). 1
- (iii) Using your answers to parts (i) and (ii) above , find an approximation for e correct to 4 decimal places. 1

### QUESTION TWO

- (a) Differentiate  $\tan(x^2)$ . 1
- (b) (i) Differentiate  $(x \sin^{-1} \frac{x}{4} + \sqrt{16-x^2})$  and simplify. 3
- (ii) Hence evaluate  $\int_0^4 \sin^{-1} \frac{x}{4} dx$ . 2
- (c) (i) Given that (0,0) is the only stationary point of  $y = x \sin^{-1} \frac{x}{4}$ , justify whether it is a minimum or maximum turning point. 2

- (ii) Hence , or otherwise, sketch the curve  $y = x \sin^{-1} \frac{x}{4}$ . 2

### QUESTION THREE

**MARKS**

- (a) If  $x = a + b$  and  $y = a - b$ ,
- (i) Show that  $\cos x + \cos y = 2 \cos\left(\frac{x+y}{2}\right) \cos\left(\frac{x-y}{2}\right)$ . 2
- (ii) Hence find the general solutions to  $\cos 4\theta + \cos 3\theta + \cos 2\theta + \cos \theta = 0$ . 4
- (b) The region in the first quadrant bounded by the curve  $y = 2 \tan^{-1} x$  and the  $y$ -axis between  $y = 0$  and  $y = \frac{\pi}{2}$  is rotated through one complete revolution about the  $y$ -axis. Find the exact volume of the solid of revolution so formed. 4

### QUESTION FOUR

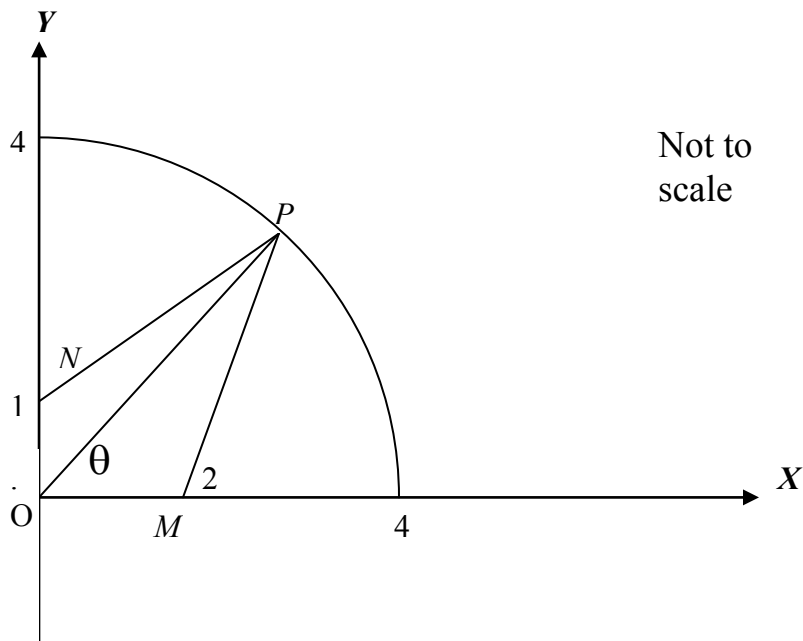
- (a) Evaluate  $\int_0^{\frac{\pi}{6}} \sin^2 x \cos^2 x \, dx$ . 3
- (b) A function is defined as  $f(x) = 1 - \cos \frac{x}{2}$ , where  $0 \leq x \leq a$ . 2
- (i) Find the largest value of  $a$  for which the inverse function  $f^{-1}(x)$  exists.
- (ii) Find the equation of  $f^{-1}(x)$ . 1
- (iii) Sketch the graph of  $y = f^{-1}(x)$ . 2
- (iv) Find the area enclosed between the curve  $y = f^{-1}(x)$ , the  $x$ -axis and  $x = 2$ . 2

**QUESTION FIVE****MARKS**

(a) Find the value of  $\lim_{x \rightarrow 0} \frac{x^2}{1 - \cos 2x}$ .

**3**

(b)



The diagram shows the part of the circle  $x^2 + y^2 = 16$  that lies in the first quadrant. The point  $P(x, y)$  is on the circle with centre at the origin O.

$M$  is on the  $x$ -axis at  $x = 2$  and  $N$  is on the  $y$ -axis at  $y = 1$ .

The size of angle  $MOP$  is  $\theta$  radians.

(i) Show that the area,  $A$ , of the quadrilateral  $OMPN$  is given by  
 $A = 4\sin\theta + 2\cos\theta$ .

**2**

(ii) Find the value of  $\tan\theta$  for which  $A$  is maximum.

**3**

- (iii) Hence determine in surd form the coordinates of  $P$  for which  $A$  is maximum. 2

### QUESTION SIX

MARKS

The straight line  $y = mx + b$  meets the parabola  $x^2 = 4ay$  at the points  $P(2ap, ap^2)$  and  $Q(2aq, aq^2)$ .

- (i) Find the equation of the chord  $PQ$  and hence, or otherwise, show that  $pq = \frac{-b}{a}$ . 3
- (ii) Prove that  $p^2 + q^2 = 4m^2 + \frac{2b}{a}$ . 2
- (iii) Given that the equation of the normal to the parabola at  $P$  is  $x + py = 2ap + ap^3$  and that  $N$ , the point of intersection of the normals at  $P$  and  $Q$ , has co-ordinates  $[-apq(p+q), a(2 + p^2 + pq + q^2)]$ , express these co-ordinates in terms of  $a, m, b$ . 2
- (iv) Suppose that the chord  $PQ$  is free to move while maintaining a fixed gradient. Find the locus of  $N$  and show that this locus is a straight line. 3

Verify that this line is a normal to the parabola.

**END OF PAPER**

Solutions to T4 3u 2006

1a)  $\frac{(3x+5)^9}{27} + C$  #

ii)  $-\frac{1}{2} \int \frac{du}{\sqrt{u}}$   $u = 3-x^2$   
 $du = -2x dx$   
 $= -\frac{1}{2} \frac{u^{\frac{1}{2}}}{\frac{1}{2}}$   
 $= -\sqrt{3-x^2} + C$  #

b)  $\sqrt{12} \sin x + 2 \cos x = A \cos x \cos x - A \sin x \sin x$   
 $A = \sqrt{(\sqrt{12})^2 + 2^2} = \sqrt{16} = 4$   $A = 4$   
 $-4 \sin x = \sqrt{12}$   $2 = 4 \cos x$   
 $\sin x < 0$   $\cos x > 0$   $x$  is in 4th Quad  
 $\tan x = -\frac{\sqrt{12}}{2} = -\sqrt{3}$   $x = \frac{5\pi}{3}$   
 $\sqrt{12} \sin x + 2 \cos x = 4 \cos(x + \frac{5\pi}{3})$  #

c) 

|         |   |       |     |
|---------|---|-------|-----|
| x       | 1 | 1 1/2 | 2   |
| y = 1/x | 1 | 2/3   | 1/2 |

Area  $\div \frac{2-1}{6} [1 + 4 \times \frac{2}{3} + \frac{1}{2}] = \frac{1}{6} (\frac{3}{3} + \frac{8}{3} + \frac{1}{2})$   
 $= \frac{25}{36}$

ii) Exact Area  $= \int_1^2 \frac{1}{x} dx = [ln x]_1^2 = ln 2 - ln 1$   
 $= ln 2$

iii)  $ln 2 \div \frac{25}{36}$   
 $e^{\frac{25}{36}} = 2$   $e = 2^{\frac{36}{25}} = 2.7132$  (4dp)

2a)  $2x \sec(x^2)$

b)  $x \cdot \frac{\frac{1}{4}}{\sqrt{1-\frac{x^2}{16}}} + \sin^{-1} \frac{x}{4} + \frac{-2x}{2\sqrt{16-x^2}}$   
 $= \frac{\frac{1}{4}x}{\sqrt{16-x^2}} + \sin^{-1} \frac{x}{4} - \frac{x}{\sqrt{16-x^2}}$   
 $= \sin^{-1} \frac{x}{4}$

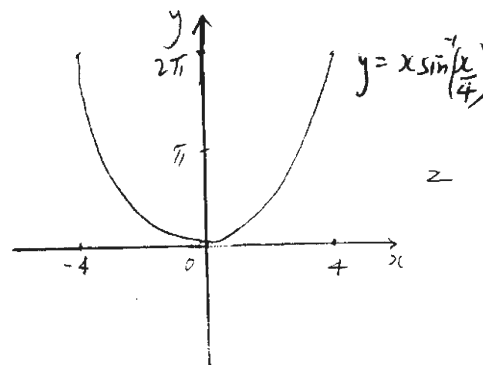
ii)  $\int_0^4 \sin^{-1} \frac{x}{4} dx = [x \sin^{-1} \frac{x}{4} + \sqrt{16-x^2}]_0^4$   
 $= 4 \sin^{-1} 1 + 0 - 0 - \sqrt{16} = 2\pi - 4$

c)  $y' = \frac{2x}{\sqrt{16-x^2}} + \sin^{-1} \frac{x}{4}$  (from b)

|       |      |   |     |
|-------|------|---|-----|
| x     | -0.1 | 0 | 0.1 |
| y'    | -0.5 | 0 | 0.5 |
| shape | ↘    | — | ↗   |

Since y is continuous and differential for  $x \leq 1$  and  $y'$  changes signs (- to +) from  $x = -0.1$  to  $x = 0.1$

$\therefore$  min turning point at  $(0, 0)$



3a)  $\cos x + \cos y = \cos(a+b) + \cos(a-b)$   
 $= \cos a \cos b - \sin a \sin b + \cos a \cos b + \sin a \sin b$   
 $= 2 \cos a \cos b = 2 \cos(\frac{x+y}{2}) \cos(\frac{x-y}{2})$   
 since  $\begin{cases} x = a+b \\ y = a-b \end{cases} \Rightarrow \begin{cases} a = \frac{x+y}{2} \\ b = \frac{x-y}{2} \end{cases}$

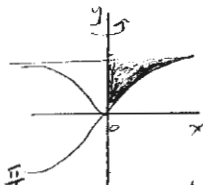
ii)  $(\cos 4\theta + \cos 2\theta) + (\cos 3\theta + \cos \theta) = 0$   
 $2 \cos 3\theta \cos \theta + 2 \cos 2\theta \cos \theta = 0$  (by p)  
 $2 \cos \theta (\cos 3\theta + \cos 2\theta) = 0$   
 $2 \cos \theta (2 \cos \frac{5\theta}{2} \cos \frac{\theta}{2}) = 0$   
 $4 \cos \theta \cos \frac{5\theta}{2} \cos \frac{\theta}{2} = 0$

$\cos \theta = 0 \Rightarrow \theta = \pm \frac{\pi}{2} + 2n\pi \quad n \in \mathbb{Z}$   
 $\cos \frac{5\theta}{2} = 0 \Rightarrow \frac{5\theta}{2} = \pm \frac{\pi}{2} + 2n\pi \Rightarrow \theta = \pm \frac{\pi}{5} + \frac{4n\pi}{5}$   
 $\cos \frac{\theta}{2} = 0 \Rightarrow \frac{\theta}{2} = \pm \frac{\pi}{2} + 2n\pi \Rightarrow \theta = \pm \pi + 4n\pi$

$\therefore$  general solns are:  
 $\theta = \pm \frac{\pi}{2} + 2n\pi, \pm \frac{\pi}{5} + \frac{4n\pi}{5}, \pm \pi + 4n\pi \quad n \in \mathbb{Z}$

$$3b) y = 2 \tan^{-1} x \quad x = \tan \frac{y}{2}$$

$$\begin{aligned} \text{Vol of revol} &= \pi \int_0^{\frac{\pi}{2}} x^2 dy \\ &= \pi \int_0^{\frac{\pi}{2}} \tan^2\left(\frac{y}{2}\right) dy = \pi \int_0^{\frac{\pi}{2}} (\sec^2 \frac{y}{2} - 1) dy \\ &= \left[ \pi \left( \tan \frac{y}{2} \right) 2 - \pi y \right]_0^{\frac{\pi}{2}} = 2\pi \tan \frac{\pi}{4} - \frac{\pi}{2} \\ &= 2\pi - \frac{\pi}{2} \text{ unit}^3 \end{aligned}$$



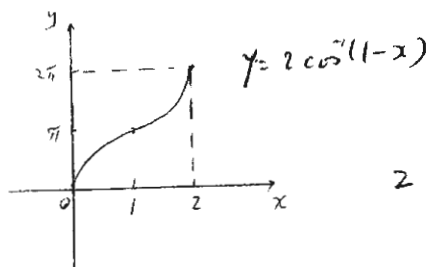
$$\begin{aligned} 4a) \int_0^{\frac{\pi}{6}} \left( \frac{1}{2} \sin 2x \right) dx &= \frac{1}{4} \int_0^{\frac{\pi}{6}} \frac{1 - \cos 4x}{2} dx \\ &= \left[ \frac{x}{8} - \frac{\sin 4x}{8 \cdot 4} \right]_0^{\frac{\pi}{6}} = \frac{\pi}{48} - \frac{\sin \frac{2\pi}{3}}{32} \\ &= \frac{\pi}{48} - \frac{\sqrt{3}}{64} \end{aligned}$$

$$b) y = 1 - \cos \frac{x}{2}$$

$$\text{Period: } \frac{2\pi}{\frac{1}{2}} = 4\pi$$

i) Largest  $a$  is  $2\pi$   
 ii)  $f^{-1}(x)$  exists for  $0 \leq x \leq 2\pi$

$$\begin{aligned} \text{ii) For inverse: } x &= 1 - \cos \frac{y}{2} \\ \cos \frac{y}{2} &= 1 - x \\ \frac{y}{2} &= \cos^{-1}(1-x) \\ y &= 2 \cos^{-1}(1-x) \end{aligned}$$



$$\begin{aligned} \text{iv) Area} &= \frac{1}{2} \times \text{Area of Rectangle} \\ &= \frac{1}{2} \times 2 \times 2\pi = 2\pi \text{ units}^2 \end{aligned}$$

Alternatively

$$\begin{aligned} \text{Area} &= 4\pi - \int_0^{\frac{\pi}{2}} (1 - \cos \frac{y}{2}) dy \\ &= 4\pi - \left[ y - 2 \sin \frac{y}{2} \right]_0^{\frac{\pi}{2}} \\ &= 4\pi - (2\pi - 0 - 0 + 0) = 2\pi \text{ units}^2 \end{aligned}$$

Q5

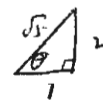
$$a) \lim_{x \rightarrow 0} \frac{x}{2 \sin x} = \lim_{x \rightarrow 0} \frac{1}{2} \left( \frac{x}{\sin x} \right) = \frac{1}{2} \times 1 = \frac{1}{2}$$

$$b) \text{Area } \triangle OPN = \frac{1}{2} \times 1 \times 4 \cos \theta = 2 \cos \theta$$

$$\text{Area } \triangle OPM = \frac{1}{2} \times 2 \times 4 \sin \theta = 4 \sin \theta$$

$$\text{Area OMPN(A)} = 4 \sin \theta + 2 \cos \theta$$

$$\begin{aligned} \text{ii) } A' &= 4 \cos \theta - 2 \sin \theta = 0 \quad 0 < \theta < \frac{\pi}{2} \\ 4 \cos \theta &= 2 \sin \theta \\ \tan \theta &= 2 \end{aligned}$$



$$A'' = -4 \sin \theta - 2 \cos \theta = \frac{-4 \times 2}{\sqrt{5}} - \frac{2}{\sqrt{5}} < 0$$

Since  $A, A', A''$  are continuous and differentiable for  $0 < \theta < \frac{\pi}{2}$  and only 1 TP  $\therefore \tan \theta = 2$  gives max  $A$

$$\begin{aligned} \text{iii) } P &= (4 \cos \theta, 4 \sin \theta) = \left( \frac{4}{\sqrt{5}}, \frac{4 \times 2}{\sqrt{5}} \right) \\ &= \left( \frac{4}{\sqrt{5}}, \frac{8}{\sqrt{5}} \right) \end{aligned}$$

Q6

$$\begin{aligned} \text{(i) slope } PQ &= \frac{a(q^2 - p^2)}{2a(q - p)} = \frac{a(q + p)(q - p)}{2a(q - p)} \\ &= \frac{q + p}{2} \end{aligned}$$

$$\text{Eq of PQ} \quad y - ap^2 = \left( \frac{q + p}{2} \right) (x - 2ap)$$

$$y = \frac{x}{2} (p + q) - apq - ap^2 + ap^2$$

$$y = \frac{x}{2} (p + q) - apq$$

At P and Q:  $y = mx + b$  and  $y = \frac{x}{2}(p+q) - apq$   
 $mx + b = \frac{x}{2}(p+q) - apq$   
 $m = \frac{p+q}{2} \quad b = -apq \quad \therefore pq = \frac{-b}{a} \quad \# \quad 1$

ii) Since  $p+q = 2m$ ,  $pq = -\frac{b}{a}$   
 $p^2 + q^2 = (p+q)^2 - 2pq = (2m)^2 - 2(-\frac{b}{a}) = 4m^2 + \frac{2b}{a} \quad \# \quad 1$

iii)  $x = -apq(p+q) = -a(-\frac{b}{a})2m \quad \therefore x = 2mb \quad \# \quad 1$   
 $y = a(2 + p^2 + pq + q^2) = a(2 + (p+q)^2 - pq) = a(2 + 4m^2 - \frac{-b}{a})$   
 $y = a(2 + 4m^2 + \frac{b}{a}) = 2a + 4am^2 + b \quad \# \quad 1$

iv) Since  $b = \frac{2c}{2m} \quad \therefore y = 2a + 4am^2 + \frac{x}{2m}$

1 Gradient of chord PQ is fixed  $\therefore m$  is a constant  
 but  $a$  is also a constant  $\therefore$  this is an equation of a straight line in gradient-intercept form.

$$x - 2my = -4am - 8am^3$$

1  $x + (-2m)y = 2a(-2m) + a(-2m)^3$

Compare this equation with equation of normal to parabola at 1

$$x + py = 2ap + ap^3$$

$p$  has been replaced by  $(-2m)$

$\therefore$  locus of N is a straight line which is normal to the parabola at point  $(2ax - 2m, a(-2m)^2)$

1 is at  $(-4m, 4am^2)$